

MUJICA: Multi-skill Unified Joint Integration of Control Architecture for Wheeled-Legged Robots

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Project Page: <https://hyzenthlayer.github.io/mujica/>

Abstract—Wheeled-legged robots hold promise for traversing complex terrains and offer superior mobility compared to legged robots. However, wheeled-legged robots must effectively balance both wheeled driving and legged control. Furthermore, due to noisy proprioceptive sensing and real-world motor constraints, realizing robust and adaptive locomotion at peak performance of motors remains challenging. We propose the Multi-skill Unified Joint Integration of Control Architecture (MUJICA), a unified, fully proprioceptive control framework for wheeled-legged robots that integrates diverse low-level skills—including omnidirectional moving, high platform climbing, and fall recovery—within a single policy. All skills, distinguished by unique indicator variables, are trained jointly with accurate DC-motor constraint modeling. Additionally, a high-level skill selector is learned to dynamically choose the optimal skill based solely on proprioceptions, enabling adaptive responses to the surrounding environment. Therefore, MUJICA enhances sim-to-real robustness and enables seamless transitions across diverse locomotion modes, facilitating autonomous adjustment to the environment. We validate our framework in both simulation and real-world experiments on the Unitree Go2-W robot, demonstrating significant improvements in adaptability and task success in unstructured environments.

I. INTRODUCTION

With significant advances in motor torque and computational power of hardware, legged robots have achieved impressive progress in traversing challenging environments. Their ability to adapt to uneven terrains and recover from perturbations makes them suitable for real-world tasks such as inspection and disaster response [1]. Despite these advances, designing control algorithms for legged robots remains challenging. Controllers must simultaneously achieve motion diversity, adaptability to different terrains, and safe deployment under real-world actuation constraints. A variety of approaches have been proposed and have demonstrated robustness in tasks such as stair climbing, slope traversal, and perturbation recovery [2]–[6]. However, owing to inherent limitations of the robot’s structure, legged locomotion is often limited in efficiency and speed on flat ground, and lacks the capability of traversing certain challenging terrains, such as high platforms.

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Fig. 1: Real-world demonstrations of MUJICA on the wheeled-legged robot. **Top row**: automatic skill switching during continuous locomotion. **Middle row**: execution of three representative skills, including high platform climbing, stair ascent, and fall recovery. **Bottom row**: snapshots of the robot climbing onto a 1-m-high platform. The results highlight MUJICA’s ability to perform diverse skills and seamlessly transition between them without external perception.

Wheeled-legged robots offer a promising alternative by combining the efficiency of wheels with the legs’ ability to cross obstacles. This hybrid design enables rapid traversal on smooth surfaces while preserving the ability to handle rough terrains and obstacles [7]. Nevertheless, the full potential of wheeled-legged robots in complex real-world environments has yet to be realized because most existing control frameworks for wheeled-legged robots are modified from legged locomotion methods [7], [8], and thus inherit their limitations.

We identify three key limitations in current legged and wheeled-legged locomotion approaches. First, most multi-skill blind controllers based solely on proprioception can only handle either a narrow range of similar skills with limited motion diversity, such as walking on flat or inclined terrains or skills with relatively low difficulty, such as climbing stairs and slopes [2]–[4], [9]. Second, for handling dissimilar skills with distinct dynamics, such as galloping, ambling and crawling, controllers often rely on manual switching [10], [11]. Moreover, these methods are primarily designed to showcase multi-skill learning paradigms, rather than to adapt to diverse and complex terrains. Third, in

MUJICA：轮腿机器人的多技能统一联合控制架构

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摘要——轮腿机器人在穿越复杂地形方面具有广阔前景，并且相比腿式机器人具有更优越的机动性。然而，轮腿机器人必须有效平衡轮式驱动和腿部控制。此外，由于本体感觉存在噪声以及真实世界中的电机约束，在电机峰值性能下实现鲁棒自适应运动仍然具有挑战性。我们提出了多技能统一联合控制架构（MUJICA），这是一种面向轮腿机器人的统一的全本体感觉控制框架，将多种低级技能——包括全向移动、高台攀爬和跌倒恢复——集成于单一策略之中。所有技能通过独特的指示变量进行区分，并在精确的直流电机约束建模下进行联合训练。此外，我们还学习了一个高层技能选择器，仅基于本体感觉动态选择最优技能，从而实现对周围环境的自适应响应。因此，MUJICA增强了仿真到现实的鲁棒性，并实现了不同运动模式之间的无缝切换，促进了对环境的自主调整。我们在宇树 Go2-W 机器人上进行了仿真和真实世界实验验证，结果表明在非结构化环境中适应性和任务成功率均得到了显著提升。

I. 引言

随着电机扭矩和硬件计算能力的显著进步，腿式机器人在穿越具有挑战性的环境方面取得了令人瞩目的进展。它们适应不平坦地形和从扰动中恢复的能力使其适用于巡检和灾害响应等现实任务 [1]。尽管取得了这些进展，设计腿式机器人的控制算法仍然具有挑战性。控制器必须同时实现运动多样性、对不同地形的适应性以及在真实驱动约束下的安全部署。已有多种方法被提出，并在爬楼梯、斜坡穿越和扰动恢复等任务中展示了鲁棒性 [2]–[6]。然而，由于机器人结构的固有局限性，腿部运动在平坦地面上的效率和速度往往受限，并且缺乏穿越某些具有挑战性地形（如高台）的能力。

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图1: MUJICA在轮腿机器人上的真实世界演示。顶行：连续运动过程中的自动技能切换。中间行：执行三种代表性技能，包括高台攀爬、上楼梯和跌倒恢复。底行：机器人攀上1米高平台的快照。结果凸显了MUJICA执行多种技能并在无需外部感知的情况下无缝切换技能的能力。

轮腿机器人通过结合轮子的高效性与腿部跨越障碍物的能力，提供了一种有前景的替代方案。这种混合设计使得在平坦表面上能够快速移动，同时保留了应对崎岖地形和障碍物 [7] 的能力。然而，轮腿机器人在复杂真实环境中的全部潜力尚未得到充分发挥，因为大多数现有的轮腿机器人控制框架都是从腿式运动方法 [7], [8], 修改而来，因此继承了它们的局限性。

我们识别出当前腿部运动和轮腿运动方法中的三个关键局限性。首先，大多数仅基于本体感觉的多技能盲控制器只能处理范围狭窄的相似技能，运动多样性有限，例如在平坦或倾斜地形上行走，或处理难度相对较低的技能，如爬楼梯和斜坡 [2]–[4], [9]。其次，对于处理具有不同动力学的异质技能，如疾驰、缓行和爬行，控制器通常依赖手动切换 [10], [11]。此外，这些方法主要旨在展示多技能学习范式，而非适应多样化和复杂的地形。第三，在

order to ensure safe deployment in the real world, limitations on robot joints are of vital importance. In the context of reinforcement learning (RL), these physical limitations are often transformed into action penalties or incorporated as constraints in optimization problems [12]–[14]. However, most learning-based methods apply simplified actuation constraints, ignoring the velocity-and-position-dependent torque limitations of direct current (DC) motors. Methods like [15] simply model the relationship between desired torques and measured torques. The lack of precise motor constraints limits both the exploitation of motor performance and the safety of sim-to-real transfer. Consequently, these issues limit the performance and safety of wheeled-legged locomotion on challenging terrains.

To address these issues, we present the Multi-skill Unified Joint Integration of Control Architecture (MUJICA) for wheeled-legged robots, a proprioception-driven control framework that unifies multiple locomotion skills—including omnidirectional moving, high platform climbing, and fall recovery—within a single network. A central feature of MUJICA is its emphasis on safety-aware learning and reliable sim-to-real transfer. We explicitly model the relationship between motor velocity and maximum torque as hard DC-motor constraints to prevent unsafe behaviors during deployment and allow the robot to exploit its motors more effectively during challenging maneuvers. Furthermore, we incorporate a high-level selector that leverages proprioceptive feedback to automate skill transitions, enabling smooth and adaptive switching across tasks without external perception.

The main contributions of this work are as follows:

- We propose a unified control architecture that jointly learns diverse and challenging skills, including omnidirectional locomotion, high platform climbing, and fall recovery, within a single blind policy.
- We present a safety-aware learning framework based on hard DC-motor constraints to ensure successful sim-to-real transfer.
- We train a high-level skill selector to automate skill transfer based on proprioceptions and the unified velocity tracking rewards.
- We validate the effectiveness of MUJICA through extensive simulation and real-world experiments on the Unitree Go2-W robot.

II. RELATED WORK

A. Learning-Based Blind Control for Legged Robots

Since only a few studies focus on wheeled-legged locomotion and most resemble legged control, this section mainly concentrates on legged locomotion. Blind control methods rely solely on proprioceptive feedback, such as joint states and IMU data, without exteroceptive sensors. Early works demonstrate end-to-end deep RL policies for quadrupeds wandering on flat indoor surfaces [16]. Policies are trained in simulation and deployed zero-shot on real robots. To improve the performance on unstructured terrains, privileged information is widely applied during policy training. RMA [17] leverages an adaptation module to mimic

a privileged base policy, while DreamWaQ [4] implements an asymmetric actor-critic to better train the policy through a critic with privileged observations. Another line of work estimates unobservable states through history observations, enabling better terrain adaptation. DreamWaQ [4] and HIM [18] both estimate the base linear velocity and a latent state vector to strengthen the robustness of locomotion, and FR-Net [5] predicts the robot collision probability and mass distribution to avoid undesirable collisions and enable deployment on diverse robots. Nevertheless, these methods do not incorporate a unified embodiment state estimation that captures diverse robot-specific information, limiting their applicability to multi-skill policy deployment.

On the other hand, policies that perform well in simulation do not necessarily guarantee safe real-world performance due to discrepancies in environment and robot dynamics. Therefore, enforcing physical constraints during training is essential to ensure safe and reliable deployment. Recent advances in constrained reinforcement learning have yielded several innovative approaches to address the challenge of policy optimization under constraints. To enhance computational tractability and simplify reward design, some approaches reformulate the traditionally constrained policy iteration process into an equivalent unconstrained optimization formulation [12], [13]. From another perspective, CaT [19] ingeniously reinterprets hard constraints as probabilistic termination mechanisms, achieving robust constraint satisfaction while maintaining computational efficiency. However, studies such as Dadiotis et al. [20] and ALARM [21] only constrain the joint torques by their maximum allowable values, neglecting the fact that the maximum torque output of a DC motor decreases at high speeds, while [15] merely models the relationship between desired torques and measured torques. This oversight limits the robot’s safety performance under extreme operating conditions. For wheeled-legged robots, exploiting extreme motor capabilities in coordination with wheel-leg coordination is essential to maximize performance, further highlighting the need for accurate torque modeling.

B. Multitask Learning in Robotics

The goal of multitask learning in robotics is to enable robots to accomplish different tasks or acquire various skills through a unified policy. Although [4], [18] enable robots to traverse stairs and flat terrains, the action spaces of these skills do not differ substantially. Therefore, multitask frameworks are required to handle extensively different tasks and enable robots to learn different skills. Existing frameworks mainly fall into two types: teacher-student policy distillation and hierarchical reinforcement learning. Teacher-student policy distillation first trains multiple expert policies, each specialized in a single task, and then uses supervised learning to train a student policy to imitate these experts [22], [23]. Hierarchical reinforcement learning also adopts a two-stage training scheme where multiple low-level single-skill expert policies are first trained. For high-level policy, some studies apply mixture-of-expert (MOE) framework to achieve multi-skill learning such as fall recovery, quadrupedal walk-

为了确保在真实世界中的安全部署，机器人关节的限制至关重要。在强化学习（RL）的背景下，这些物理限制通常被转化为动作惩罚或作为约束纳入优化问题 [12]–[14]。然而，大多数基于学习的方法采用简化的驱动约束，忽略了直流（DC）电机的速度和位置相关的力矩限制。像 [15]这样的方法仅简单地对期望力矩和测量力矩之间的关系进行建模。缺乏精确的电机约束既限制了电机性能的发挥，也限制了仿真到现实迁移的安全性。因此，这些问题限制了轮腿运动在具有挑战性的地形上的性能和安全性。

为了解决这些问题，我们提出了用于轮腿机器人的多技能统一联合控制架构（MUJICA），这是一种本体感觉驱动的控制框架，在单一网络内统一了多种运动技能——包括全向移动、高台攀爬和跌倒恢复。MUJICA的一个核心特点是强调安全感知学习和可靠的仿真到现实迁移。我们明确地将电机速度与最大力矩之间的关系建模为硬性直流电机约束，以防止部署过程中的不安全行为，并允许机器人在具有挑战性的机动中更有效地利用其电机。此外，我们引入了一个高层选择器，利用本体感觉反馈自动实现技能转换，从而在没有外部感知的情况下实现跨任务的平滑自适应切换。

本研究的主要贡献如下：

- 我们提出了一种统一控制架构，在单一盲策略内联合学习多样且具有挑战性的技能，包括全向运动、高台攀爬和跌倒恢复。
- 我们提出了一种基于硬直流电机约束的安全感知学习框架，以确保从仿真到真实世界的成功迁移。
- 我们训练了一个高层技能选择器，基于本体感觉和统一的速度跟踪奖励来自动化技能迁移。
- 我们通过在宇树Go2-W机器人上的广泛仿真和真实世界实验验证了MUJICA的有效性。

II. 相关工作

A. 基于学习的足式机器人盲控制 机器人

由于仅有少数研究聚焦于轮腿运动，且大多数与腿部控制相似，本节主要关注腿部运动。盲控制方法仅依赖本体感觉反馈，如关节状态和IMU数据，而不使用外部传感器。早期工作展示了四足机器人在平坦室内表面上漫游的端到端深度强化学习策略 [16]。策略在仿真中训练，并在真实机器人上零样本部署。为了提升在非结构化地形上的表现，特权信息在策略训练中被广泛应用。RMA [17]利用自适应模块来模仿

特权基础策略，而DreamWaQ [4] 实现了非对称演员-评论家架构，通过具有特权观测的评论家更好地训练策略。另一类工作通过历史观测估计不可观测状态，从而实现更好的地形适应。DreamWaQ [4] 和HIM[18] 均估计基座线速度和潜在状态向量以增强运动的鲁棒性，FR-Net [5]则预测机器人碰撞概率和质量分布，以避免不良碰撞并支持在多样化机器人上的部署。然而，这些方法未纳入能够捕获多样化机器人特定信息的统一具身状态估计，限制了其在多技能策略部署中的适用性。

另一方面，在仿真中表现良好的策略并不一定能保证真实世界中的安全性能，因为环境和机器人动力学存在差异。因此，在训练过程中强制执行物理约束对于确保安全可靠的部署至关重要。约束强化学习的最新进展已经产生了多种创新方法，以应对在约束条件下进行策略优化的挑战。为了增强计算可处理性并简化奖励设计，一些方法将传统的约束策略迭代过程重新表述为等效的无约束优化公式 [12], [13]。从另一个角度来看，CaT [19] 巧妙地将硬约束重新解释为概率性终止机制，在保持计算效率的同时实现了稳健的约束满足。然而，诸如Dadiotis等人 [20] 和ALARM [21] 的研究仅通过最大允许值来约束关节力矩，忽略了直流电机的最大扭矩输出在高速时会下降的事实，而 [15] 仅建模了期望力矩与测量力矩之间的关系。这种疏忽限制了机器人在极端运行条件下的安全性能。对于轮腿机器人而言，利用极限电机能力并与轮腿协调相结合对于最大化性能至关重要，这进一步凸显了精确力矩建模的必要性。

B. 机器人多任务学习

机器人多任务学习的目标是通过统一策略使机器人能够完成不同任务或获得各种技能。尽管 [4], [18] 使机器人能够穿越楼梯和平坦地形，但这些技能的动作空间并无实质性差异。因此，需要多任务框架来处理差异巨大的任务，并使机器人能够学习不同的技能。现有框架主要分为两类：师生策略蒸馏和分层强化学习。师生策略蒸馏首先训练多个专家策略，每个策略专精于单一任务，然后使用监督学习训练学生策略以模仿这些专家 [22][23]。分层强化学习也采用两阶段训练方案，首先训练多个低级单技能专家策略。对于高级策略，一些研究应用混合专家（MOE）框架来实现多技能学习，如跌倒恢复、四足行走、

IV. METHOD

In this section, we present the detailed methodology of our proposed MUJICA framework. The overall architecture is illustrated in Fig. 2.

A. State Estimator

Relying solely on proprioceptions, the robot cannot directly access terrain geometry or contact conditions, which are crucial for skill execution. Thus, to infer the nearby environment, inspired by HIM [18], which constructs an internal model to predict system responses to disturbances, we develop a state estimator based on a buffer of past observations. Aiming to capture temporal dependencies, the estimator utilizes an online encoder to process the past $H = 6$ frames of observations through a GRU-based network. In addition to estimating the latent state vector $\hat{e}_t$ and the robot base linear velocity $\hat{v}_t$, we also predict collision probabilities $\hat{c}_t$ for each robot component and the wheel-ground distance $\hat{u}_t$. Explicitly predicting velocity helps the robot comprehend the locomotion dynamics and distinguish between stable and unstable locomotion regimes. The wheel-ground distance estimation reflects terrain roughness, allowing the robot to suppress unnecessary leg lifting on flat ground and better coordinate wheel-leg interaction. Inferring collision probabilities helps the robot detect contact with obstacles, such as using head collisions to sense upcoming ledges for climbing, and enables terrain adaptive recovery behaviors.

The estimation process can be formulated as follows:

$$\mathbf{f}_t = \text{GRU}(\text{NN}(\mathbf{o}_{t-H:t}), \mathbf{f}_{t-1}) \quad (5)$$

where $\mathbf{f}_t = [\hat{v}_t, \hat{c}_t, \hat{u}_t, \hat{e}_t]^T$, while NN indicates fully connected layers.

On the purpose of effectively training the online encoder, we introduce a reference encoder that maps the successor observation $\mathbf{o}_{t+1}$ to a latent state e_t as the supervision target, and a contrastive learning loss is applied.

To summarize, the estimator loss can be formulated as:

$$\mathcal{L}^{\text{Estimate}} = \mathcal{L}^{\text{Pred}} + \mathcal{L}^{\text{SwAV}}(e_t, \hat{e}_t) \quad (6)$$

$$\mathcal{L}^{\text{Pred}} = \mathcal{L}_{\text{MSE}}(\mathbf{v}_t, \hat{\mathbf{v}}_t) + \mathcal{L}_{\text{BCE}}(\mathbf{c}_t, \hat{\mathbf{c}}_t) + \mathcal{L}_{\text{MSE}}(\mathbf{u}_t, \hat{\mathbf{u}}_t) \quad (7)$$

where $\mathcal{L}_{\text{MSE}}$ equals mean squared error loss, $\mathcal{L}_{\text{BCE}}$ equals binary cross-entropy loss, and SwAV is the contrastive learning loss following [27].

B. Asymmetric Actor-Critic Framework

As formalized in Section III-B, the wheeled-legged robot operates under partial observability during deployment, yet requires safety assessment based on unobservable terrain properties. Meanwhile, the critic exposed to full state information learns the value function much faster, better guiding the learning of the actor [28]. Therefore, MUJICA proposes an asymmetric actor-critic framework. The policy network π_θ receives as input the history-embedded observation $\mathbf{f}_t$ from the online encoder and the current observation $\mathbf{o}_t$.

In parallel, to address the constrained optimization problem in Equation (1), we adopt the P3O [12] framework, which employs a reward critic

and a constraint critic. Both critics are

conditioned on privileged observations $\mathbf{o}_t^{\text{priv}}$ to separately evaluate the state value and the constraint violation.

P3O transforms the constrained optimization problem into an unconstrained objective using a penalty-based iterative approach.

Consequently, P3O loss can be formulated as:

$$\mathcal{L}^{\text{P3O}}(\theta) = \mathcal{L}_R^{\text{CLIP}}(\theta) + \kappa \sum_{i=1}^k \max\{0, \mathcal{L}_{C_i}^{\text{CLIP}}(\theta)\} \quad (8)$$

$$\mathcal{L}_R^{\text{CLIP}}(\theta) = \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi}} [-\min\{r(\theta)A_R^\pi(s, a), \\ \text{clip}(r(\theta), 1 - \epsilon, 1 + \epsilon)A_R^\pi(s, a)\}] \quad (9)$$

$$\mathcal{L}_{C_i}^{\text{CLIP}}(\theta) = \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi}} [\max\{r(\theta)A_{C_i}^\pi(s, a), \text{clip}(r(\theta), \\ 1 - \epsilon, 1 + \epsilon)A_{C_i}^\pi(s, a)\} + (1 - \gamma)(J_{C_i}(\pi) - \delta_i)] \quad (10)$$

where $r(\theta)$ is the importance sampling ratio between the new and old policies, A_R^π and $A_{C_i}^\pi$ are the reward and i th constraint advantages, $d^\pi(s)$ is the discounted future state distribution, and κ is the penalty factor. $\mathcal{L}_R^{\text{CLIP}}$ is the clipped PPO surrogate objective [29], ensuring stable reward-driven updates, while $\mathcal{L}_{C_i}^{\text{CLIP}}$ encodes constraint-specific losses with clipping to prevent overly aggressive updates. $(J_{C_i}(\pi) - \delta_i)$ penalizes expected constraint costs exceeding δ_i , scaled by $(1 - \gamma)$. Summed over all constraints, the P3O objective maximizes reward while suppressing unsafe behaviors, yielding policies that are both performant and safe. To enable the robot to gradually acquire higher-difficulty skills, we design a curriculum learning scheme to train the actor-critic network (see Section V-A for details).

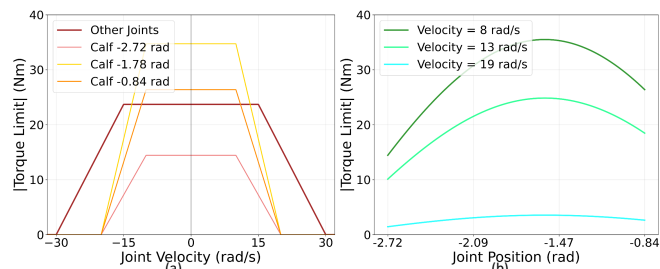


Fig. 3: (a) The velocity-dependent torque limit of all joints. Calf joint limits are also influenced by joint position. (b) The position-based torque limit of calf joints.

C. Multi-Task Reward and Constraint

On the purpose of learning various skills within a single policy, we propose the skill indicator ζ_t . Each skill is associated with a unique indicator variable so that the observation spaces of individual skills are disentangled along this specific dimension. Consequently, through the skill indicator ζ_t , our proposed architecture can be extended to various wheeled-legged tasks. In this study, we design three representative and challenging tasks: (i) omnidirectional moving on ordinary terrains, (ii) high platform climbing, and (iii) fall recovery. The omnidirectional locomotion skill requires tracking linear

IV. 方法

在本节中，我们详细介绍了所提出的MUJICA框架的方法。整体架构如图2所示。

A. 状态估计器

仅依靠本体感觉，机器人无法直接获取地形几何或接触条件，而这些对于技能执行至关重要。因此，为了推断周围环境，受HIM [18]，构建内部模型以预测系统对扰动响应的启发，我们开发了一种基于过去观测缓冲区的状态估计器。为了捕捉时间依赖性，该估计器利用在线编码器通过基于GRU的网络处理过去的 $H = 6$ 帧观测。除了估计潜在状态向量 $\hat{e}_t$ 和机器人基座线速度 $\hat{v}_t$ 外，我们还预测每个机器人组件的碰撞概率 $\hat{c}_t$ 以及轮地距离 $\hat{u}_t$ 。显式预测速度有助于机器人理解运动动力学，并区分稳定与不稳定的运动状态。轮地距离估计反映了地形粗糙度，使机器人能够在平坦地面上抑制不必要的抬腿动作，并更好地协调轮腿交互。推断碰撞概率有助于机器人检测与障碍物的接触，例如利用头部碰撞感知即将到来的边缘以进行攀爬，并实现地形自适应的恢复行为。

估计过程可以表述如下：

$$\mathbf{f}_t = \text{GRU}(\text{NN}(\mathbf{o}_{t-H:t}), \mathbf{f}_{t-1}) \quad (5)$$

其中 $\mathbf{f}_t = [\hat{v}_t, \hat{c}_t, \hat{u}_t, \hat{e}_t]^T$ ，而 NN 表示全连接层。

为了有效训练在线编码器，我们引入了一个参考编码器，将后继观测 $\mathbf{o}_{t+1}$ 映射到潜在状态 e_t 作为监督目标，并应用对比学习损失。

综上所述，估计器损失可以表述为：

$$\mathcal{L}^{\text{Estimate}} = \mathcal{L}^{\text{Pred}} + \mathcal{L}^{\text{SwAV}}(e_t, \hat{e}_t) \quad (6)$$

$$\mathcal{L}^{\text{Pred}} = \mathcal{L}_{\text{MSE}}(\mathbf{v}_t, \hat{\mathbf{v}}_t) + \mathcal{L}_{\text{BCE}}(\mathbf{c}_t, \hat{\mathbf{c}}_t) + \mathcal{L}_{\text{MSE}}(\mathbf{u}_t, \hat{\mathbf{u}}_t) \quad (7)$$

其中 $\mathcal{L}_{\text{MSE}}$ 等于均方误差损失， $\mathcal{L}_{\text{BCE}}$ 等于二元交叉熵损失，而 SwAV 是遵循 [27] 的对比学习损失。

B. 非对称演员-评论家框架

如第III-B节所述，轮腿机器人在部署过程中面临部分可观测性，但仍需基于不可观测的地形属性进行安全评估。与此同时，能够获取完整状态信息的评论家网络学习价值函数的速度要快得多，从而能更好地引导演员网络 [28] 的学习。因此，MUJICA提出了一种非对称演员-评论家框架。策略网络 π_θ 接收来自在线编码器的历史嵌入观测 $\mathbf{f}_t$ 和当前观测 $\mathbf{o}_t$ 作为输入。

同时，为了解决公式(1)中的约束优化问题，我们采用P3O [12] 框架，该框架使用奖励评论家网络

和约束评判器。这两个评论家均以特权观测 $\mathbf{o}_t^{\text{priv}}$ 为条件，分别评估状态值和约束违反情况。

P3O通过基于惩罚的迭代方法，将约束优化问题转化为无约束目标。

因此，P3O损失可以表示为：

$$\mathcal{L}^{\text{P3O}}(\theta) = \mathcal{L}_R^{\text{CLIP}}(\theta) + \kappa \sum_{i=1}^k \max\{0, \mathcal{L}_{C_i}^{\text{CLIP}}(\theta)\} \quad (8)$$

$$\mathcal{L}_R^{\text{CLIP}}(\theta) = \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi}} [-\min\{r(\theta)A_R^\pi(s, a), \\ \text{clip}(r(\theta), 1 - \epsilon, 1 + \epsilon)A_R^\pi(s, a)\}] \quad (9)$$

$$\mathcal{L}_{C_i}^{\text{CLIP}}(\theta) = \mathbb{E}_{\substack{s \sim d^\pi \\ a \sim \pi}} [\max\{r(\theta)A_{C_i}^\pi(s, a), \text{clip}(r(\theta), \\ 1 - \epsilon, 1 + \epsilon)A_{C_i}^\pi(s, a)\} + (1 - \gamma)(J_{C_i}(\pi) - \delta_i)] \quad (10)$$

其中 $r(\theta)$ 是新旧策略之间的重要性采样比率， A_R^π 和 $A_{C_i}^\pi$ 分别是奖励优势和 i 第个约束优势， $d^\pi(s)$ 是折扣未来状态分布， κ 是惩罚因子。 $\mathcal{L}_R^{\text{CLIP}}$ 是裁剪的PPO替代目标 [29]，确保基于奖励的更新稳定进行，而 $\mathcal{L}_{C_i}^{\text{CLIP}}$ 则编码了带有裁剪的约束特定损失，以防止过于激进的更新。 $(J_{C_i}(\pi) - \delta_i)$ 对超过 δ_i 的期望约束成本进行惩罚，并按 $(1 - \gamma)$ 进行缩放。对所有约束求和后，P3O目标在最大化奖励的同时抑制不安全行为，从而得到既高效又安全的策略。为了使机器人能够逐步获得更高难度的技能，我们设计了一种课程学习方案来训练演员-评论家网络（详见第V-A节）。

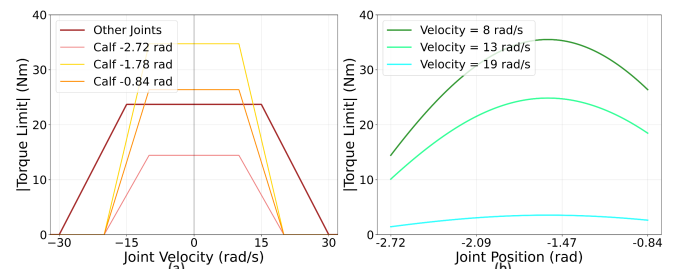


图3: (a) 所有关节的速度相关扭矩限制。小腿关节极限也受关节位置影响。(b) 小腿关节的基于位置的扭矩限制。

C. 多任务奖励与约束

为了在单一策略中学习多种技能，我们提出了技能指示器 ζ_t 。每个技能都与一个独特的指示变量相关联，使得个体技能的观测空间沿这一特定维度得以解耦。因此，通过技能指示器 ζ_t ，我们提出的架构可以扩展到各种轮腿式任务中。在本研究中，我们设计了三个具有代表性和挑战性的任务：(i) 在普通地形上的全向移动，(ii) 高台攀爬，以及 (iii) 跌倒恢复。全向运动技能需要跟踪线

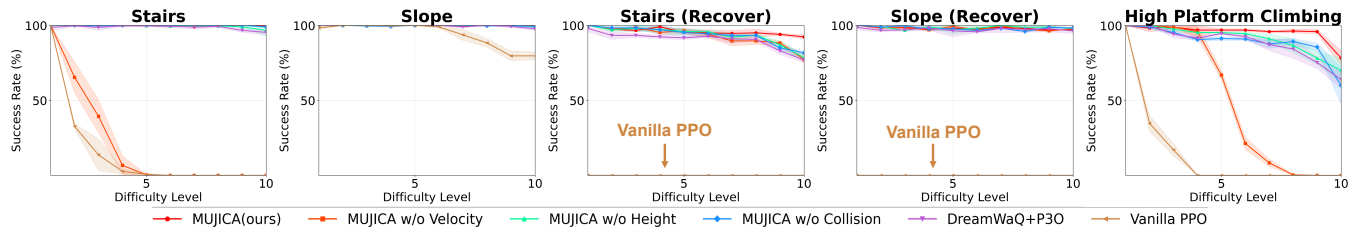


Fig. 4: Success rate of different algorithms on all difficulty levels of representative tasks.

and angular velocities, while the high platform climbing skill only needs to track linear velocity. For fall recovery, the robot needs to learn to flip itself over from arbitrary poses.

TABLE I: Primary rewards and constraints of MUJICA

Reward R_t or Constraint C_t	Formula	Applicable Tasks
$R_{cmd_v, t}$	$\exp(-\ \mathbf{cmd}_{xy, t} - \mathbf{v}_{xy, t}\ ^2 / \sigma^2)$	i, ii
$R_{cmd_\omega, t}$	$\exp(-\ \mathbf{cmd}_{\omega, t} - \omega_{z, t}\ ^2 / \sigma^2)$	i
$R_{gravity, t}$	$\exp(-\angle(\mathbf{g}_t^b, \mathbf{g}^{world}) / \sigma^2)$	iii
$R_{poserr, t}$	$\exp(-\ \mathbf{q} - \mathbf{q}_{stand}\ ^2 / \sigma^2)$ if $ \angle(\mathbf{g}_t^b, \mathbf{g}^{world}) < \epsilon$	iii
$C_{DC-motor, t}$	$\sum_{i=1}^{16} \mathbf{1}_{ \tau_t^i \geq \tau_{limit}^i}$	i, ii, iii
$C_{collision, t}$	$\sum_i c_t^i, i = \text{thigh, calf}$	i, ii

The primary and task-relevant rewards and constraints are listed in TABLE I, while others are supplemented in the video. Notably, collision constraints are imposed on the thigh and calf to encourage the robot to interact with the environment using wheels rather than legs. Moreover, to account for the physical limitations of electric motors, we introduce the DC-motor constraint to all the tasks. Specifically, inspired by the motor operating regions from [30], shown in Fig. 3, each actuator is subject to a velocity-dependent torque limit, where the maximum torque remains constant at low speeds but decreases linearly with increasing velocity at high speeds. The calf joints connect the thigh and calf links, so the torque is influenced by both joint velocity and joint position, with the maximum torque exhibiting a cosine dependence on the joint position at fixed velocity, whereas other joint motors are solely affected by joint velocity. The specific values are from the official motor manual of Unitree.

D. Skill Selector

After training several robust locomotion skills (Training S1), to enable adaptive behavior across diverse terrains, we adopt a proprioception-based high-level skill selector that automates the activation of low-level locomotion policies, i.e., chooses the appropriate ζ_t automatically. In particular, the high-level selector takes the last H observations $\mathbf{o}_{t-H:t}$, excluding ζ_t , as input, and outputs a probability distribution over the predefined set of locomotion skills. An asymmetric actor-critic network is implemented to train the skill selector on top of the frozen skill policies through the unified velocity tracking rewards (Training S2). This hierarchical design

simplifies low-level learning and improves generalization through flexible skill composition.

V. EXPERIMENTS

We train MUJICA on diverse terrains and conduct extensive experiments in both simulation and the real world to validate its effectiveness and peak performance. In detail, the following experiments aim to address:

- Can MUJICA effectively complete diverse tasks, and how does its performance compare with baseline methods and ablated variants? (In Section V-B)
- Does the proposed skill selector enable reliable switching across tasks with large variations in characteristics, leading to improved sequential-task success rates? (In Section V-C)
- Can MUJICA policies be transferred to real wheeled-legged robots, demonstrating zero-shot sim-to-real performance? (In Section V-D)

A. Training

1) *Settings*: All experiments in simulation are conducted using the IsaacLab platform with 4,096 parallel environments on a single NVIDIA RTX 4090 GPU. We choose the Unitree Go2-W shown in Fig. 2 to validate the proposed architecture. The low-level controller and the high-level selector both work at a frequency of 50 Hz, while the physics simulation is executed at 200 Hz. The low-level training process is carried out for 30,000 iterations, and the high-level training is conducted for 10,000 iterations. In the fall recovery task, each episode lasts 6 s, including 2 s of free fall from the air with a random posture and 4 s for recovery, during which the task will not terminate. For the other tasks, each episode lasts 20 s and terminates when the base collision occurs.

2) *Curriculum Learning*: To enable the robot to gradually explore motions that adapt to challenging terrains, following [31], we apply a curriculum learning framework.

We construct a 33×20 grid-shaped multi-task environment, in which 4,096 robots are assigned to different rows, each corresponding to a specific terrain and its corresponding task, while each column represents a curriculum difficulty level.

During Training S1, we explicitly set the correct skill indicator according to the task assigned to each robot. For both omnidirectional locomotion and fall recovery, the terrains are set as stairs, slopes, discretized terrains and rough terrains. For high platform climbing, the terrain is set as a large depression on the ground that requires the

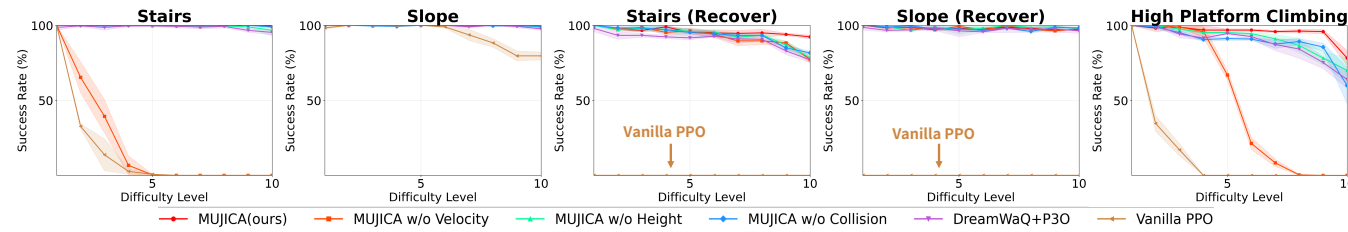


图4: 不同算法在代表性任务的不同算法在所有难度等级的代表性表现
所有难度等级上的成功

代表性任务。

速度和角速度，而高台攀爬技能仅需跟踪线速度。对于跌倒恢复，机器人需要学习从任意姿态翻转起身。

简化了底层学习，并通过灵活的技能组合提升了泛化能力。

V. 实验

我们在多样地形上训练MUJICA，并在仿真和真实世界中开展广泛实验，以验证其有效性和峰值性能。具体而言，以下实验旨在解决：

- MUJICA能否有效完成多样化任务，其性能与基线方法和消融变体相比如何？（见第V-B节）
- 所提出的技能选择器能否在特性差异较大的任务间实现可靠切换，从而提升顺序任务成功率？（见第V-C节）
- MUJICA策略能否迁移到真实轮腿机器人上，展示零样本仿真到现实迁移的性能？（见第V-D节）

A. 训练

1) 设置：所有仿真实验均在IsaacLab平台上进行，使用单个NVIDIA RTX 4090 GPU运行4,096个并行环境。我们选择图2所示的宇树Go2-W来验证所提出的架构。底层控制器和高层选择器均以50 Hz的频率工作，而物理仿真以200 Hz执行。底层训练过程进行30,000次迭代，高层训练进行10,000次迭代。在跌倒恢复任务中，每个回合持续6秒，包括2秒从空中以随机姿态自由落体和4秒用于恢复，期间任务不会终止。对于其他任务，每个回合持续20秒，并在发生基座碰撞时终止。

2) 课程学习：为了使机器人能够逐步探索适应具有挑战性的地形的运动，按照[31]，我们采用了课程学习框架。

我们构建了一个 33×20 网格状多任务环境，其中4,096个机器人被分配到不同的行，每一行对应一个特定的地形及其相应的任务，而每一列代表一个课程难度等级。

在训练阶段S1中，我们根据分配给每个机器人的任务明确设置正确的技能指示器。对于全向运动和跌倒恢复，地形设置为楼梯、斜坡、离散地形和崎岖地形。对于高台攀爬，地形设置为地面上的大型凹陷，要求

表I: MUJICA的主要奖励与约束

奖励 R_t 或约束 C_t	公式	适用 任务
$R_{cmd_v, t}$	实验 $(-\ \mathbf{cmd}_{xy, t} - \mathbf{v}_{xy, t}\ ^2 / \sigma^2)$	i, ii
$R_{cmd_\omega, t}$	实验 $(-\ \mathbf{cmd}_{\omega, t} - \omega_{z, t}\ ^2 / \sigma^2)$	i
$R_{gravity, t}$	实验 $(-\angle(\mathbf{g}_t^b, \mathbf{g}^{world}) / \sigma^2)$	iii
$R_{poserr, t}$	$\exp(-\ \mathbf{q} - \mathbf{q}_{stand}\ ^2 / \sigma^2)$ 如果 $ \angle(\mathbf{g}_t^b, \mathbf{g}^{world}) < \epsilon$	iii
$C_{DC-motor, t}$	$\sum_{i=1}^{16} \mathbf{1}_{ \tau_t^i \geq \tau_{限制}^i}$	i, ii, iii
$C_{collision, t}$	$\sum_i c_t^i, i = \text{大腿, 小腿}$	i, ii

主要且与任务相关的奖励和约束列于表I中，其余内容在视频中补充。值得注意的是，对大腿和小腿施加了碰撞约束，以鼓励机器人使用轮子而非腿部与环境交互。此外，考虑到电机物理限制，我们在所有任务中引入了直流电机约束。具体而言，受 [30], 图3所示电机工作区域的启发，每个执行器都受到速度相关扭矩限制，其中最大扭矩在低速时保持恒定，但在高速时随速度增加而线性减小。小腿关节连接大腿和小腿连杆，因此扭矩同时受关节速度和关节位置的影响，在固定速度下，最大扭矩对关节位置表现出余弦依赖，而其他关节电机仅受关节速度影响。具体数值来自宇树官方电机手册。

D. 技能选择器

在训练了多种稳健的运动技能（训练阶段S1）之后，为了在多种地形上实现自适应行为，我们采用了一种基于本体感觉的高层技能选择器，它能够自动激活底层运动策略，即自动选择合适的 ζ_t 。具体而言，高层选择器将最近的 H 个观测 $\mathbf{o}_{t-H:t}$ （不包括 ζ_t ）作为输入，并输出在预定义运动技能集合上的概率分布。我们实现了非对称演员-评论家网络，通过统一速度跟踪奖励（训练阶段S2）在冻结技能策略之上训练技能选择器。这种分层设计

robot to climb out and enables climbing in any direction. Considering the objectives of these three skills, for omnidirectional locomotion and high platform climbing, the curriculum progresses once the robot successfully maintains velocity tracking for a period of time, and regresses if the robot’s traveled distance is far below the theoretical distance. For fall recovery, the curriculum progresses when the robot is within a certain range of the standard standing posture, and regresses otherwise. The corresponding terrains are as follows with the terrain level $l \in [1, 20]$:

- Stairs: with step width of 0.3 m and step height of $0.05 + 0.18 \times l/20$ m;
- Slope: with slope of $0.5 \times l/20 \times 100\%$;
- Discretized terrain: with 30 rectangular bumps on the ground whose widths range from 1 m to 2 m and heights given by $0.05 + 0.17 \times l/20$ m;
- Rough terrain: with height sampled uniformly from $[0.04, 0.12]$ m;
- Pit: a rectangular pit on the ground with dimensions $2 \text{ m} \times 4 \text{ m}$ (length $\times$ width) and depth $0.05 + l/20$ m.

During Training S2, all environments are treated equally, with the expectation that the robots will learn to select skills on their own to continuously track the velocity commands.

TABLE II: Randomized Simulation Parameters

Parameter	Randomization Range
Static Friction	$[0.6, 1.0]$
Dynamic Friction	$[0.4, 0.8]$
Base Mass Bias	$[-1.0, 3.0]$ kg
External Force	Every 2-3 s at $[-10.0, 10.0]$ N
Push Robots	Every 8-12 s at $[-1.0, 1.0]$ m/s on x or y direction
Motor Gain Multiplier	$[0.8, 1.2]$

3) *Domain Randomizations*: To improve the robustness of our proposed method, and to ensure smooth sim-to-real transfer, we widely implement domain randomizations during training. TABLE II lists the randomized variables and their uniformly sampled ranges.

B. Comparison and Ablation Experiments

1) *Baseline Comparison and State Estimator Ablation*: We first compare our method with several baselines and ablations on the separate skill performance as follows:

- MUJICA (Ours): Training with all modules.
- MUJICA w/o Velocity: Training without base linear velocity estimation.
- MUJICA w/o Wheel Height: Training without wheel-ground distance estimation.
- MUJICA w/o Collision: Training without robot segment collision estimation.
- DreamWaQ [4] + P3O [12]: A framework trained with constraints, which explicitly estimates base linear velocity and implicitly infers privileged state using the context-aided VAE estimator network.

- Vanilla PPO [29]: Training with only proprioception.

All the above methods are trained under the same terrain curriculum and reward function. To assess policy effectiveness and robustness, 5 representative tasks are chosen, and 10 increasing difficulty levels are designed, with level gaps doubled relative to training. Success for omnidirectional locomotion is maintaining a 1 m/s target velocity for 20 s; for recovery, achieving a stable standing pose similar to the nominal posture in 4 s; and for high platform climbing, climbing onto the platform within 3 s and maintaining stability. All methods are tested with 4 random seeds and mean success rates are reported alongside the standard deviation.

The results are demonstrated in Fig. 4. Firstly, all methods except vanilla PPO are able to recover from slopes at all difficulties since rolling allows the robot to exploit external forces on smooth slope surfaces. In terms of overall performance, our MUJICA framework utilizing base velocity, wheel-ground distance, and collision estimations consistently outperforms all baselines and ablations across all tasks. Notably, the performance gap widens significantly at higher difficulty levels, demonstrating the effectiveness of our proposed architecture in enhancing robustness. The ablation results indicate that removing any of the three estimation components leads to a noticeable decline in performance, highlighting that accurate evaluation of the surrounding environment requires the joint prediction of all three elements. Moreover, omitting velocity estimation is particularly detrimental for non-recovery tasks, highlighting its importance for adaptation to dynamic environments.

Compared to DreamWaQ [4] + P3O [12], which also employs an estimation framework with constraints, our method demonstrates superior performance. This suggests that our specific choice of estimation targets and network architecture more effectively captures the necessary information for diverse tasks in wheeled-legged robots. It should be noted that, due to the remarkable mobility of wheeled-legged robots, even policies trained with the simplest PPO baseline algorithm can successfully accomplish the slope-traversing task. However, PPO fails to learn any fall recovery skills due to the absence of state estimation.

2) *DC-Motor Constraint Ablation*: To further investigate the role of the DC-motor constraint, we analyze the torque–velocity distributions of the thigh joints across diverse terrains in simulation. As shown in Fig. 5, in the unconstrained case (Figs. 5a–d), the joints exhibit severe violations, with over 90% of the samples lying outside the feasible motor region. For clarity, the plots are truncated to ± 100 N·m, although many outliers with higher magnitudes exist. Moreover, without the DC-motor constraint, the joints undergo persistent oscillations, and during high platform climbing, the rear thigh joints frequently hit their positive joint limits. At this limit, large torques are generated when pushing against the ground, which directly triggers motor faults on the real robot (as shown in Fig. 6). In contrast, with the DC-motor constraint enforced (Figs. 5e–h), our method effectively keeps the distributions within the feasible region, reduces violation rates to below 3.5%, and ensures stable,

机器人爬出并能够向任意方向攀爬。考虑到这三种技能的目标，对于全向运动和高台攀爬，当机器人成功维持速度跟踪一段时间后课程推进，若机器人行进距离远低于理论距离则课程回退。对于跌倒恢复，当机器人处于标准站立姿态的一定范围内时课程推进，否则课程回退。相应地形如下，地形等级为 $l \in [1, 20]$:

- 楼梯: 台阶宽度为0.3米，台阶高度为 $0.05 + 0.18 \times l/20$ 米;
- 斜坡: 坡度为 $0.5 \times l/20 \times 100\%$;
- 离散地形: 地面上有30个矩形凸起，其宽度范围为1米至2米，高度由 $0.05 + 0.17 \times l/20$ 米给定;

- 粗糙地形: 高度从 $[0.04, 0.12]$ 米均匀采样;

- 坑: 地面上的一个矩形坑，尺寸为 $2 \text{米} \times 4 \text{米}$ （长 $\times$ 宽），深度为 $0.05 + l/20$ 米。
在训练阶段S2中，所有环境均被同等对待，期望机器人能够自主学习选择技能，以持续跟踪速度指令。

表II: 随机化仿真参数

参数	随机化范围
静摩擦	$[0.6, 1.0]$
动摩擦	$[0.4, 0.8]$
基座质量偏差	$[-1.0, 3.0]$ 千克
外力	每2-3秒施加 $[-10.0, 10.0]$ 牛
推机器人	每8-12秒以 $[-1.0, 1.0]$ 米/秒沿 x 或 y 方向
电机增益倍数	$[0.8, 1.2]$

3) 域随机化：为了提高我们提出方法的鲁棒性，并确保从仿真到现实的无缝迁移，我们在训练过程中广泛实施了域随机化。表II列出了随机化变量及其均匀采样范围。

B. 对比与消融实验

1) 基线对比与状态估计器消融：我们首先将我们的方法与几种基线和消融方法在各项技能表现上进行比较，具体如下：

- MUJICA（我们的方法）：使用所有模块进行训练。
- MUJICA无速度：训练时不使用基座线速度估计。

- MUJICA无轮高：训练时不使用轮地距离估计。

- MUJICA无碰撞：训练时不使用机器人分段碰撞估计。

- DreamWaQ [4] + P3O [12]: 一种在约束条件下训练的框架，它显式估计基座线速度，并利用上下文辅助的VAE估计器网络隐式推断特权状态。

- Vanilla PPO [29]: 仅使用本体感觉进行训练。
上述所有方法均在相同的地形课程和奖励函数下进行训练。为评估策略的有效性和鲁棒性，我们选择了5个代表性任务，并设计了10个递增的难度等级，等级间隔相对于训练时加倍。全向运动的成功标准为以1 m/s的目标速度维持20秒；恢复任务的成功标准为在4秒内达到与标称姿态相似的稳定站立姿态；高台攀爬的成功标准为在3秒内攀上平台并保持稳定。所有方法均使用4个随机种子进行测试，并报告平均成功率及标准差。

结果如图4所示。首先，除Vanilla PPO外，所有方法都能在所有难度下从斜坡中恢复，因为滚动使机器人能够利用光滑斜坡表面上的外力。就整体性能而言，我们利用基座速度、轮地距离和碰撞估计的MUJICA框架在所有任务中 consistently 优于所有基线和消融变体。值得注意的是，在较高难度等级下，性能差距显著扩大，这证明了我们提出的架构在增强鲁棒性方面的有效性。消融结果表明，移除三个估计组件中的任何一个都会导致性能明显下降，凸显了对周围环境的准确评估需要三者联合预测。此外，省略速度估计对非恢复任务尤为不利，这强调了其对适应动态环境的重要性。

与同样采用带约束估计框架的DreamWaQ [4] + P3O [12], 相比，我们的方法表现出更优越的性能。这表明我们在估计目标和网络架构上的特定选择，能更有效地捕捉轮腿机器人多样化任务所需的信息。需要指出的是，由于轮腿机器人具有卓越的机动性，即使使用最简单的PPO基线算法训练的策略也能成功完成斜坡穿越任务。然而，由于缺乏状态估计，PPO无法学习任何跌倒恢复技能。

2) 直流电机约束消融：为了进一步研究直流电机约束的作用，我们在仿真中分析了不同地形下大腿关节的扭矩-速度分布。如图5所示，在无约束情况下（图5a–d），关节表现出严重的违规现象，超过90%的样本位于可行电机区域之外。为清晰起见，图中截断至 ± 100 N·m，尽管存在许多幅度更大的离群值。此外，在没有直流电机约束的情况下，关节持续振荡，在高台攀爬过程中，后大腿关节频繁触及正向关节极限。在此极限下，当推压地面时会产生大扭矩，直接触发真实机器人上的电机故障（如图6所示）。相比之下，在施加直流电机约束后（图5e–h），我们的方法有效将分布保持在可行区域内，将违规率降至3.5%以下，并确保稳定、

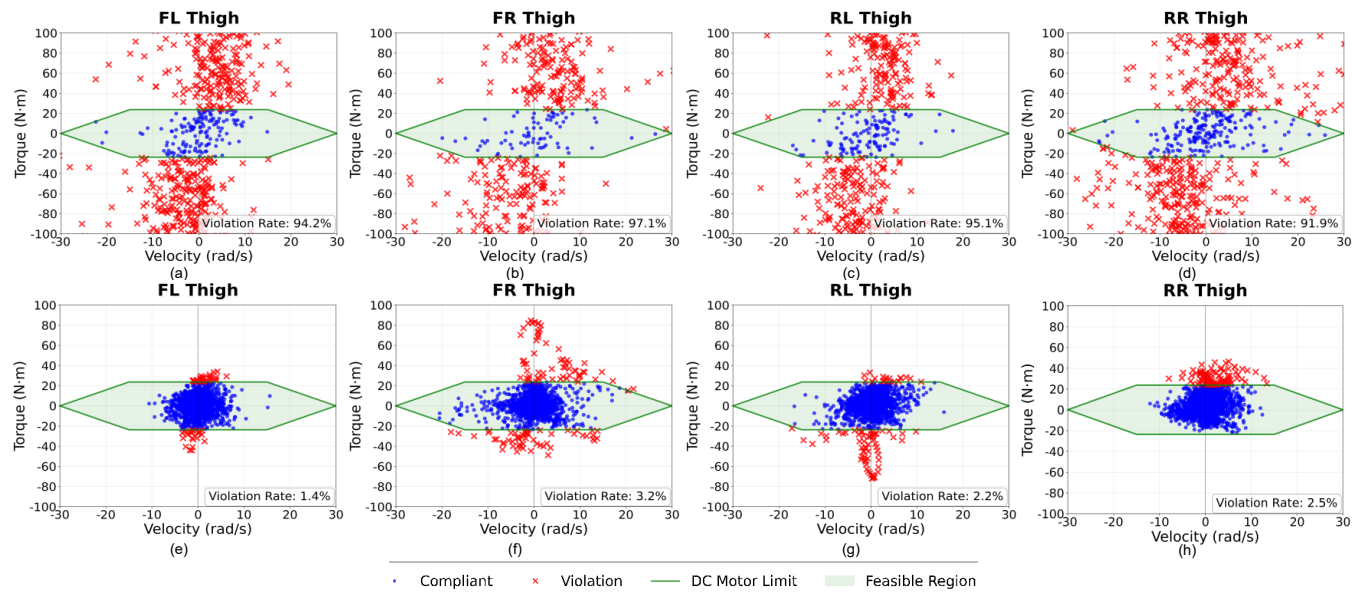


Fig. 5: Torque–velocity distributions of thigh joints across diverse terrains. Green area indicates compliance with DC-motor constraints. Blue dots are motor states within the Green area, while red crosses are not. (a)–(d) show results without DC-motor constraints, while (e)–(h) show constrained results.

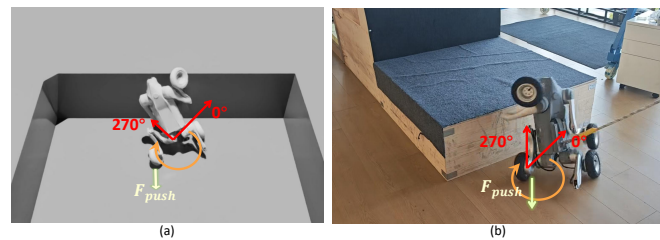


Fig. 6: High platform climbing without DC-Motor constraints. The rear thigh joint is at joint limit and will generate large torque to jump highly. (a) In simulation (b) In real world.

hardware-executable joint behaviors. It’s worth noting that the dense cluster of points at zero velocity mainly results from impacts when the robot jumps from outside into the pit, which impose large impulses on the thigh joints.

TABLE III: Stage-wise success rate on the sequential terrains

Method	Stair Recovery	Stair Climbing	High Platform Climbing
w/o indicator (shared reward)	✗	✗	✗
w/o indicator (separate reward)	72%	72%	38%
with skill selector (ours)	95%	95%	91%

C. Evaluation of Skill Selector

To validate whether the high-level skill selector can autonomously switch skills to handle complex environments, we evaluate three variants on a single episode that chains three tasks at middle difficulty (i.e., difficulty 5 in 10) in order: (i) stair recovery, (ii) stair climbing, and (iii) high platform climbing. Each run must complete all three in

sequence to be counted as success. We perform 100 episodes with identical terrains and manually provided commands. Failure is defined as base collision during stair climbing or high platform climbing, or if the task objective cannot be achieved within 3 seconds after adjusting the command.

We compare our method with the following two baselines:

- w/o indicator (shared reward): Training without the skill indicator and with unified rewards across all tasks.
- w/o indicator (separate reward): Training without the skill indicator and with reward terms differing by tasks.

The results are presented in TABLE III. Policies without a skill indicator may confuse behaviors across tasks, thereby impairing single-task performance. In particular, the unified-reward training scheme often collapses (i.e., fails to learn skills), while the separated-reward scheme still produces lower success rates on medium-difficulty tasks compared to the proposed hierarchical control architecture. This highlights that reward engineering alone is insufficient for multi-skill integration. Without an explicit indicator, the policy struggles to disentangle task-specific strategies and generalize across heterogeneous objectives. In contrast, incorporating the skill indicator and the high-level skill selector enables clear task differentiation, stabilizes training, and ultimately supports reliable sequential execution in complex scenarios.

D. Real-World Experiments

To validate MUJICA’s zero-shot sim-to-real performance, we deploy the policies on the Unitree Go2-W robot in the real world. All onboard computations are performed on an NVIDIA Jetson Orin NX. These experiments aim to assess the robustness of individual skills and the effectiveness of the skill selector in autonomous multi-skill execution.

1) *Individual Skill Validation*: We first evaluate each skill under ordinary and challenging conditions. In stair recovery

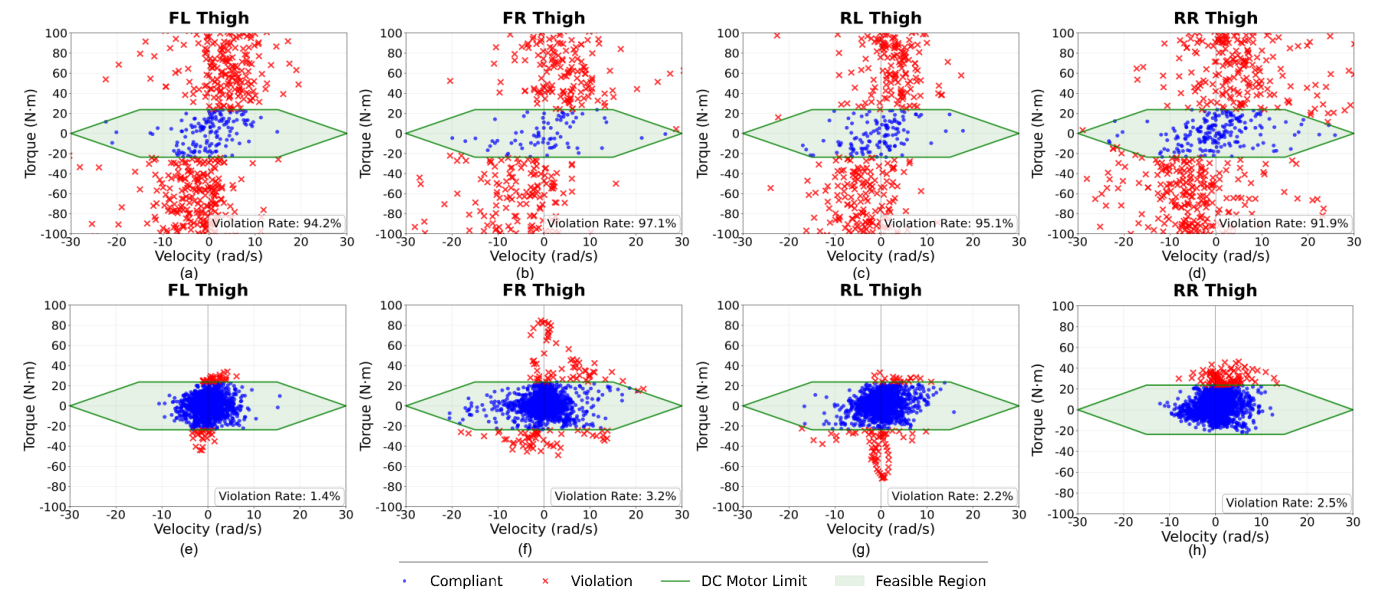


图5：多种地形下大腿关节的扭矩-速度分布。绿色区域表示符合直流电机约束。蓝色点是位于绿色区域内的电机状态，而红色叉号则不在该区域内。(a)–(d) 显示未施加直流电机约束的结果，而 (e)–(h) 显示施加约束后的结果。

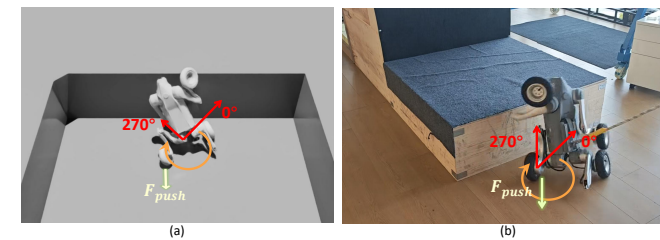


图6：无直流电机约束下的高台攀爬。后大腿关节处于关节极限，将产生较大扭矩以实现高跳。(a) 仿真中 (b) 真实世界中。

才能计为成功。我们使用相同地形和手动提供的指令执行了100个回合。失败定义为在爬楼梯或高台攀爬过程中发生基座碰撞，或在调整指令后3秒内无法达成任务目标。

我们将我们的方法与以下两种基线进行比较：

- 无指示器（共享奖励）：训练时不使用技能指示器，并在所有任务上采用统一奖励。
- 无指示器（分离奖励）：训练时不使用技能指示器，且奖励项因任务而异。

结果如表III所示。没有技能指示器的策略可能会混淆不同任务之间的行为，从而损害单任务性能。特别是，统一奖励训练方案经常崩溃（即无法学习技能），而分离奖励方案在中等难度任务上的成功率仍低于所提出的分层控制架构。这凸显了仅靠奖励工程不足以实现多技能集成。没有显式指示器，策略难以区分任务特定策略，也难以在异构目标之间进行泛化。相比之下，引入技能指示器和高层技能选择器能够实现清晰的任务区分，稳定训练过程，并最终支持在复杂场景中可靠地顺序执行。

D. 真实世界实验

为了验证MUJICA的零样本仿真到真实迁移性能，我们在真实世界中于宇树Go2-W机器人上部署了这些策略。所有有机载计算均在NVIDIA Jetson Orin NX上完成。这些实验旨在评估个体技能的鲁棒性以及技能选择器在自主多技能执行中的有效性。

1) 个体技能验证：我们首先在普通和挑战性条件下评估每项技能。在楼梯恢复

表III：连续地形上的分阶段成功率

方法	楼梯恢复	爬楼梯攀爬	高台攀爬
无指示器 (共享奖励)	✗	✗	✗
无指示器 (分离奖励)	72%	72%	38%
带技能选择器 (我们的)	95%	95%	91%

C. 技能选择器评估

为验证高层技能选择器能否自主切换技能以应对复杂环境，我们在一个单集上评估了三种变体，该单集按顺序串联了三个中等难度（即10级中的第5级）任务：

(i) 楼梯恢复，(ii) 爬楼梯，以及 (iii) 高台攀爬。每次运行必须依次完成全部三个任务，



Fig. 7: Real-world evaluations of individual skills. **Top row:** Fall recovery on stairs. **Middle row:** Ascending 20-cm-high stairs. **Bottom row:** Climbing an 80-cm-high platform.

(Fig. 7 Top Row), the robot successfully recovers from an upside-down state on 30° stairs, using calf strikes to generate torque and rotate the base, demonstrating robustness on uneven terrains at random joint positions. In stair climbing (Fig. 7 Middle Row), it steadily ascends stairs with varying riser heights and irregular edges. For high platform climbing, the robot clears an 80-cm-high outdoor platform (Fig. 7 Bottom Row) and achieves a challenging 1-m-high indoor platform (Fig. 1 Bottom Row), approaching its mechanical and actuation limit. The maneuver requires squatting the hind thigh joints to store energy, extending forelegs to hook onto the box, and lifting the hind legs in coordination, which demonstrates full wheel–leg synergy. To our knowledge, no previous work has enabled a wheeled-legged robot to accomplish this extreme task. Across all tasks, the motor protection mechanism—which will be triggered by excessive current or overheating—remains inactive, showing that the algorithm exploits the full potential of wheeled-legged locomotion.

2) *Multi-Skill Transition Tests*: We then evaluate MUJICA’s ability to autonomously integrate multiple skills during continuous real-world missions (Fig. 1 Top Row). In one demonstration, the robot starts from a random fallen posture, executes recovery, climbs a stair and a ramp, and finally climbs a 60-cm-high platform. The high-level skill selector seamlessly switches policies based on proprioceptions without human intervention: detecting the fallen posture triggers recovery, resuming the upright posture switches to omnidirectional moving, and encountering the box activates the high-platform-climbing skill. Full demonstrations of real-world experiments are available in the supplementary video.

VI. CONCLUSIONS

In this work, we introduce MUJICA, a unified control architecture that enables wheeled-legged robots to acquire and integrate multiple challenging locomotion skills within

a single proprioceptive policy. By incorporating a skill selector and a safety-aware learning framework with precise DC-motor constraints, MUJICA achieves smooth transitions across diverse tasks such as omnidirectional locomotion, high platform climbing, and fall recovery. Extensive experiments demonstrate that the proposed approach not only improves robustness compared to baseline and ablated variants, but also achieves reliable zero-shot sim-to-real transfer. These results highlight MUJICA’s potential to advance the autonomy and adaptability of wheeled-legged robots operating in complex real-world environments. Future work will explore extending the framework to a broader set of locomotion skills and enabling adaptive coordination on unstructured terrains.

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图7：个体技能的真实世界评估。顶行：在楼梯上进行跌倒恢复。中间行：攀爬20厘米高楼梯。底行：攀爬80厘米高平台。

（图7顶行）中，机器人成功从30°楼梯上的倒置状态恢复，利用小腿撞击产生扭矩并旋转基座，展示了在随机关节位置的不平坦地形上的鲁棒性。在爬楼梯（图7中间行）中，机器人稳定攀爬具有不同台阶高度和不规则边缘的楼梯。对于高台攀爬，机器人越过80厘米高的室外平台（图7底行），并完成了具有挑战性的1米高室内平台（图1底行），接近其机械与驱动极限。该操作需要下蹲后腿髋关节以储存能量，伸展前腿钩住箱子，并协调抬起后腿，充分展示了轮腿协同。据我们所知，此前尚无工作使轮腿机器人能够完成这一极限任务。在所有任务中，电机保护机制——该机制将在过电流或过热时触发——始终保持未激活状态，表明该算法充分发挥了轮腿运动的全部潜力。

2) 多技能切换测试：随后，我们评估了MUJICA在连续真实世界任务中自主整合多种技能的能力（图1顶行）。在一次演示中，机器人从随机跌倒姿态开始，执行恢复动作，攀爬楼梯和坡道，最后攀爬60厘米高平台。高层技能选择器基于本体感觉无缝切换策略，无需人工干预：检测到跌倒姿态即触发恢复，恢复直立姿态后切换至全向移动，遇到箱子则激活高台攀爬技能。真实世界实验的完整演示可在补充视频中查看。

六、结论

在本工作中，我们介绍了MUJICA，一种统一控制架构，使轮腿机器人能够在单一本体感觉策略中获取并整合多种具有挑战性的运动技能。

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